

RELATIONS

Let $A = \{1, 2\}$, $B = \{1, 2, 3\}$

$A \times B = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\}$

All subsets of $A \times B$ are relations from A to B .

Domain of a Relation = $\{a \in A \mid (a, b) \in R\}$

Range of a Relation = $\{b \in B \mid (a, b) \in R\}$

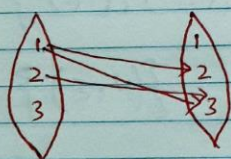
Let $R = \{(a, b) \in A \times B \mid a < b\}$

(a) Find the ordered Pairs in R : $R = \{(1, 2), (1, 3), (2, 3)\}$

(b) Find the Domain & Range in R : $D = \{1, 2\}$, $R = \{2, 3\}$

(c) Is $1R3$, $2R2$, $3R2$? I
Yes No No

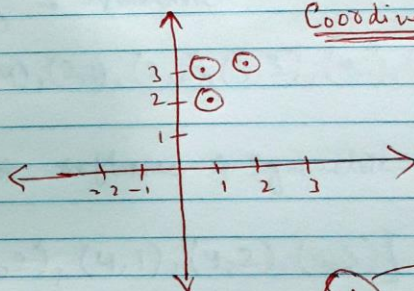
Arrow Diagram



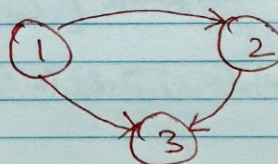
Matrix

	1	2	3
1	0	1	1
2	0	0	1
3	0	0	0

Coordinate Graph



Directed Graph



$$R = \{(1,1), (1,2), (2,1), (1,-1), (2,2)\}$$

$$R_1 = \{(a,b) \mid a \leq b\} = \{(1,1), (1,2), (2,2)\}$$

$$R_2 = \{(a,b) \mid a > b\} = \{(2,1), (1,-1)\}$$

$$R_3 = \{(a,b) \mid a = b \vee a = -b\} = \{(1,1), (1,-1), (2,2)\}$$

$$R_4 = \{(a,b) \mid a = b\} = \{(1,1), (2,2)\}$$

$$R_5 = \{(a,b) \mid a = b + 1\} = \{(2,1)\}$$

$$R_6 = \{(a,b) \mid a + b \leq 3\} = \{(1,1), (1,2), (2,1), (1,-1)\}$$

$A \times A$ is known as the universal Relation.

TYPES OF RELATION

- 1- Reflexive $\forall a \in A, \forall (a, a) \in R$
- 2- Symmetric $\forall (a, b) \in R, \text{ if } (a, b) \in R \text{ then } (b, a) \in R$
- 3- Antisymmetric $\forall (a, b) \in R, \text{ if } (a, b) \wedge (b, a) \in R \text{ then } a = b$
- 4- Transitive $\forall (a, b, c) \in R, \text{ if } (a, b) \wedge (b, c) \in R \text{ then } (a, c) \in R$
- 5- Irreflexive $\forall a \in A, \forall (a, a) \notin R$
- 6- Asymmetric $\text{Asymmetric} = \text{Irreflexive} \wedge \text{Antisymmetric}$

Examples: $A = \{1, 2, 3, 4\}$

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$$

Holds no Property

$$R_2 = \{(1, 1), (1, 2), (2, 1)\} \rightarrow \text{Symmetric}$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\} \rightarrow \text{Reflexive \& Symmetric}$$

$$R_6 = \{(3, 4)\} \rightarrow \text{not reflexive \& Symmetric}$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\} \rightarrow \text{Antisymmetric \& Irreflexive, Asymmetric}$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$$

Reflexive

$$\rightarrow \forall a \in A, \forall (a, a) \in R$$

$$R_1 = \{(1,1), (2,2), (3,3), (4,4)\}$$

	1	2	3	4
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1



Diagonal has all 1's

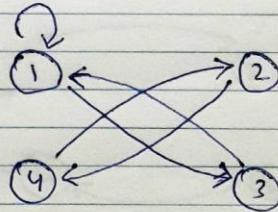
Every node has loop.

Symmetric

$$\rightarrow \forall (a,b) \in R, \text{ If } (a,b) \in R \text{ then } (b,a)$$

$$R_1 = \{(1,1), (1,3), (2,4), (3,1), (4,2)\}$$

	1	2	3	4
1	1	0	1	0
2	0	0	0	1
3	1	0	0	0
4	0	1	0	0



$$M = M^t$$

Pair of incoming & outgoing node

bidirectional links

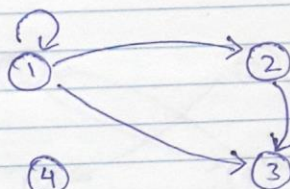
3- Transitive

$\forall a, b, c \in A, \text{ if } (a, b) \in R \wedge (b, c) \in R \text{ then } (a, c) \in R$

$$R_1 = \{(1,1), (1,2), (1,3), (2,3)\}$$

	1	2	3	4
1	1	1	1	0
2	0	0	1	0
3	0	0	0	0
4	0	0	0	0

No identification



Three nodes have indirect path than 1st & 3rd have direct path.

or Forms a triangle

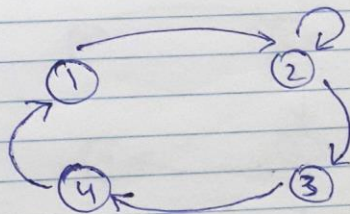
4- Antisymmetric

$\forall a, b \in A, \text{ if } ((a, b) \wedge (b, a)) \in R \text{ then } a = b \in R$

$$R_1 = \{(1,2), (2,2), (2,3), (3,4), (4,1)\}$$

	1	2	3	4
1	0	1	0	0
2	0	1	1	0
3	0	0	0	1
4	1	0	0	0

$$\begin{array}{l} M_{ij} = M_{ji} \quad i \neq j \\ M_{12} = M_{21} \\ M_{23} = M_{32} \quad | = 0 \\ 1 = 0 \end{array}$$



No Pairs of arrows between distinct node

No bidirectional link

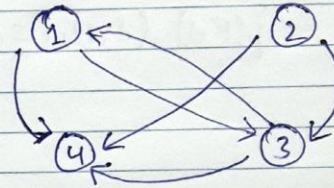
5- Irreflexive:

$$\forall a \in A, \forall (a, a) \notin R$$

$$R_1 = \{(1,3), (1,4), (2,3), (2,4), (3,1), (3,4)\}$$

	1	2	3	4
1	0	0	1	1
2	0	0	1	1
3	1	0	0	1
4	0	0	0	0

All diagonal have 0's



→ no loops

6- Asymmetric:

A
Asymmetric = Antisymmetric + Irreflexive

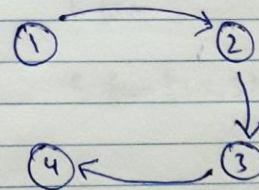
$$R = \{(1,2), (2,3), (3,4)\}$$

	1	2	3	4
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1
4	0	0	0	0

All 0's in diagonal

$$M_{ij} \neq M_{ji} \quad i \neq j$$

$$0 \neq 1$$



No bidirectional link
No loop

Combining Relations:

$$\text{let } A = \{1, 2, 3\}, B = \{1, 2, 3, 4\}$$

$$R_1 = \{(1, 1), (2, 2), (3, 3)\}$$

$$R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$$

$$R_1 \cup R_2 = \{(1, 1), (2, 2), (3, 3), (1, 2), (1, 3), (1, 4)\}$$

$$R_1 \cap R_2 = \{(1, 1)\}$$

$$R_1 - R_2 = \{(2, 2), (3, 3)\}$$

$$R_2 - R_1 = \{(1, 2), (1, 3), (1, 4)\}$$

$$R_1 \oplus R_2 = \{(1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\}$$

$$\text{SoR} = \{(1, 0), (1, 1), (2, 1), (2, 2), (3, 0), (3, 1)\}$$

$$R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\} \quad R \text{ from } \{1, 2, 3\} \text{ to } \{1, 2, 3, 4\}$$

$$S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\} \quad S \text{ from } \{1, 2, 3, 4\} \text{ to } \{0, 1, 2\}$$

Inverse of Relation

$$R^{-1} = \{(1, 1), (4, 1), (3, 2), (1, 3), (4, 3)\}$$

Equivalence relation: Reflexive, Symmetric & transitive.

Partial order relation: Reflexive, Antisymmetric & transitive.